PRP Mini Project

Project title: Employee Attrition Prediction using Machine Learning Model

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Sem: 2 Cohort 2

Objective : Build a Machine Learning model to predict employee attrition and help the company take proactive decisions to retain employees and reduce turnover costs.

Tools Used : Google Sheets → Clean employee data ● Weka → Predict who is likely to leave ● Flourish → Create visual stories for HR ● Napkin AI → Explain patterns & insights

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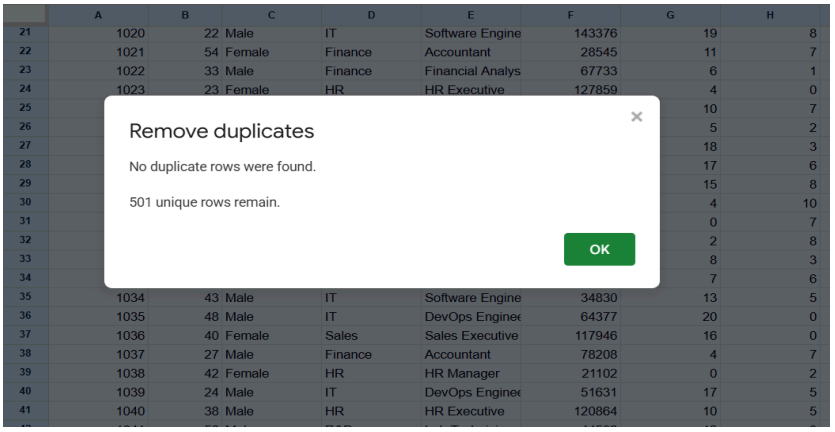
1. Load the Dataset:

- Screenshot of First 20 rows:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	EmployeeID	Age	Gender	Department	JobRole	MonthlyIncome	YearsAtCompan	YearsSinceLastF	JobSatisfaction	WorkLifeBalance	EducationLevel	TrainingHoursLa	OverTime	PerformanceRat	Attrition
2	1001	50	Female	IT	Software Engine	118694	10	7	1	1	2	74	Yes	3	Yes
3	1002	45	Male	IT	DevOps Enginee	141324	3	7	4	3	5	1	No	4	No
4	1003	51	Male	IT	Software Engine	20311	20	0	4	2	5	48	Yes	3	Yes
5	1004	49	Male	R&D	Lab Technician	17747	15	2	4	3	4	63	No	1	No
6	1005	24	Female	IT	Software Engine	84092	18	6	1	1	2	3	No	1	No
7	1006	35	Female	Sales	Sales Executive	83148	1	3	4	3	4	70	Yes	4	Yes
8	1007	29	Female	Finance	Accountant	125510	2	0	4	2	4	1	No	2	No
9	1008	25	Female	IT	DevOps Enginee	24692	17	9	4	2	2	61	Yes	4	No
10	1009	35	Female	IT	DevOps Enginee	26534	7	6	4	1	5	79	No	5	No
11	1010	39	Male	IT	Software Engine	97798	20	7	2	1	4	59	No	4	Yes
12	1011	50	Female	Sales	Sales Executive	66214	12	0	1	3	1	0	No	4	No
13	1012	29	Female	HR	HR Executive	139375	10	2	1	4	3	34	No	1	Yes
14	1013	26	Male	HR	HR Executive	149633	6	9	1	4	1	71	Yes	4	Yes
15	1014	55	Male	Finance	Financial Analys	67256	15	6	3	2	5	43	No	3	Yes
16	1015	56	Male	IT	Software Engine	128252	14	2	1	1	2	13	No	3	No
17	1016	30	Male	HR	HR Executive	146278	14	9	2	4	5	50	No	1	Yes
18	1017	53	Female	Finance	Financial Analys	102939	4	6	3	1	3	28	Yes	4	No
19	1018	34	Female	Sales	Sales Executive	77623	6	10	3	2	4	65	No	2	Yes
20	1019	27	Male	Finance	Financial Analys	110259	11	3	2	2	3	61	Yes	1	Yes

2. Clean & Prepare the Dataset:

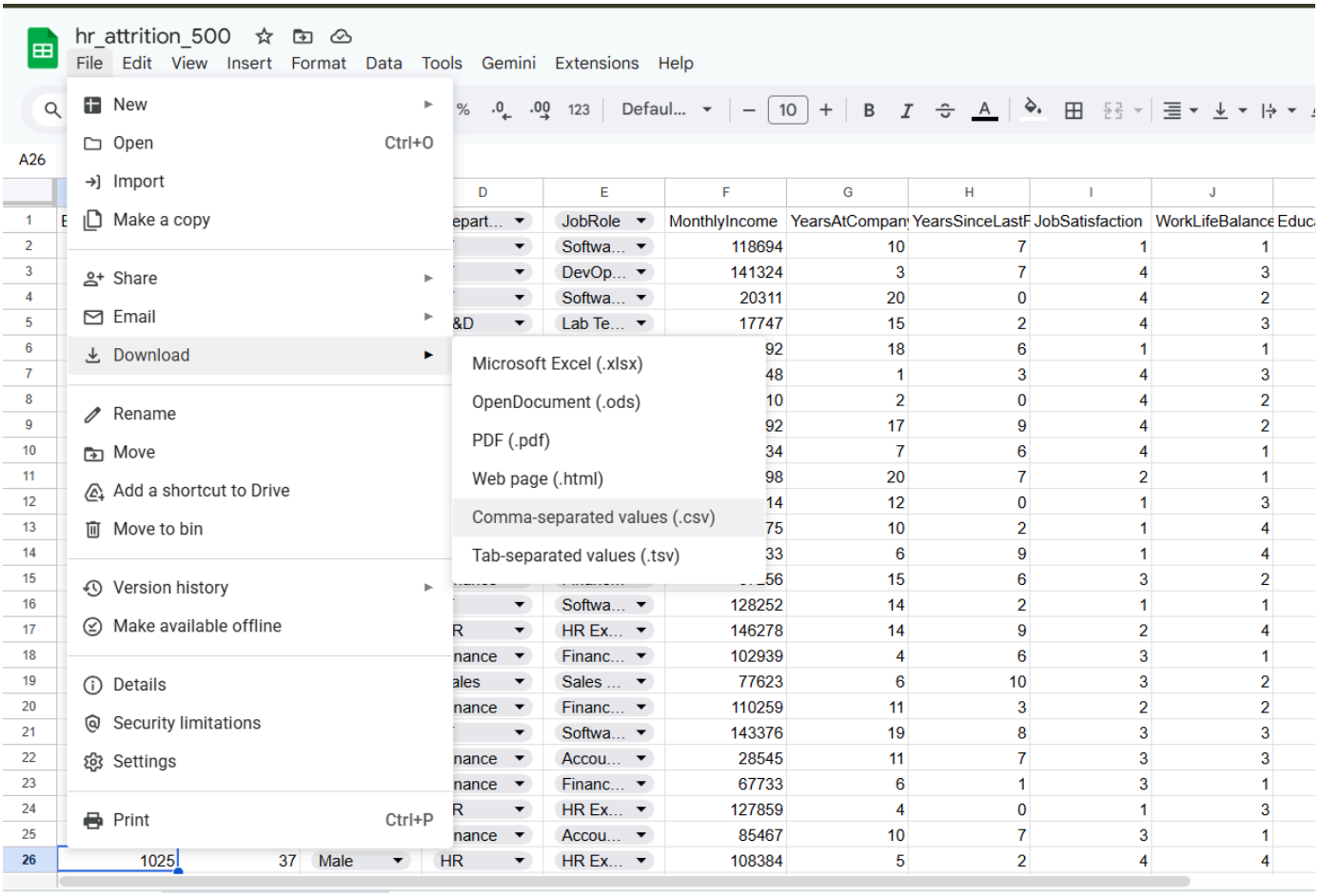
- Screenshot of duplicates removal



- Cleaned dataset

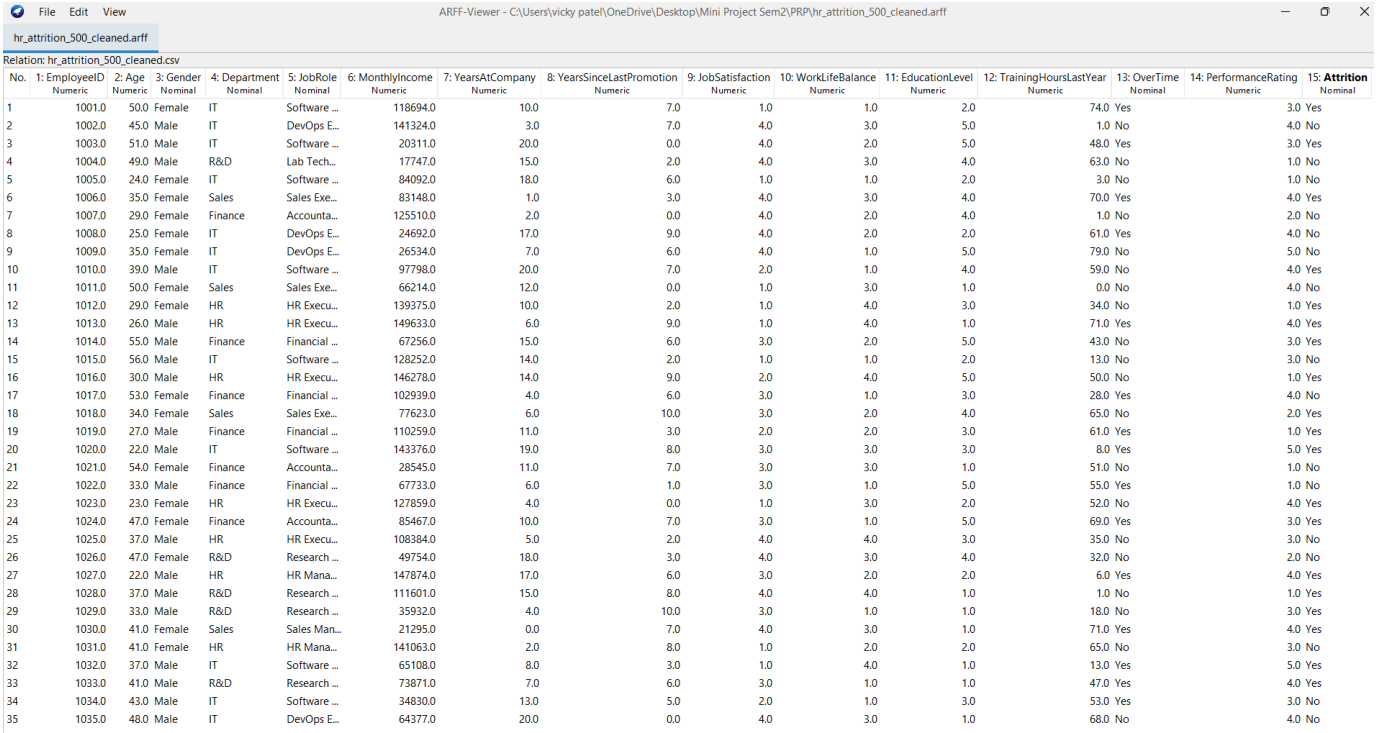
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
EmployeeID	Age	Gender	Depart...	JobRole	MonthlyIncome	YearsAtCompan	YearsSinceLastF	JobSatisfaction	WorkLifeBalance	EducationLevel	TrainingHoursLa	OverTi...	PerformanceRat	Attrition
1001	50	Female	IT	Softwa...	118694	10	7	1	1	2	74	Yes	3	Yes
1002	45	Male	IT	DevOp...	141324	3	7	4	3	5	1	No	4	No
1003	51	Male	IT	Softwa...	20311	20	0	4	2	5	48	Yes	3	Yes
1004	49	Male	R&D	Lab Te...	17747	15	2	4	3	4	63	No	1	No
1005	24	Female	IT	Softwa...	84092	18	6	1	1	2	3	No	1	No
1006	35	Female	Sales	Sales ...	83148	1	3	4	3	4	70	Yes	4	Yes
1007	29	Female	Finance	Accou...	125510	2	0	4	2	4	1	No	2	No
1008	25	Female	IT	DevOp...	24692	17	9	4	2	2	61	Yes	4	No
1009	35	Female	IT	DevOp...	26534	7	6	4	1	5	79	No	5	No
1010	39	Male	IT	Softwa...	97798	20	7	2	1	4	59	No	4	Yes
1011	50	Female	Sales	Sales ...	66214	12	0	1	3	1	0	No	4	No
1012	29	Female	HR	HR Ex...	139375	10	2	1	4	3	34	No	1	Yes
1013	26	Male	HR	HR Ex...	149633	6	9	1	4	1	71	Yes	4	Yes
1014	55	Male	Finance	Financ...	67256	15	6	3	2	5	43	No	3	Yes
1015	56	Male	IT	Softwa...	128252	14	2	1	1	2	13	No	3	No
1016	30	Male	HR	HR Ex...	146278	14	9	2	4	5	50	No	1	Yes
1017	53	Female	Finance	Financ...	102939	4	6	3	1	3	28	Yes	4	No
1018	34	Female	Sales	Sales ...	77623	6	10	3	2	4	65	No	2	Yes
1019	27	Male	Finance	Financ...	110259	11	3	2	2	3	61	Yes	1	Yes
1020	22	Male	IT	Softwa...	143376	19	8	3	3	3	8	Yes	5	Yes
1021	54	Female	Finance	Accou...	28545	11	7	3	3	1	51	No	1	No
1022	33	Male	Finance	Financ...	67733	6	1	3	1	5	55	Yes	1	No
1023	23	Female	HR	HR Ex...	127859	4	0	1	3	2	52	No	4	Yes
1024	47	Female	Finance	Accou...	85467	10	7	3	1	5	69	Yes	3	Yes

- Screenshot of “Download CSV”



3. Convert CSV to ARFF:

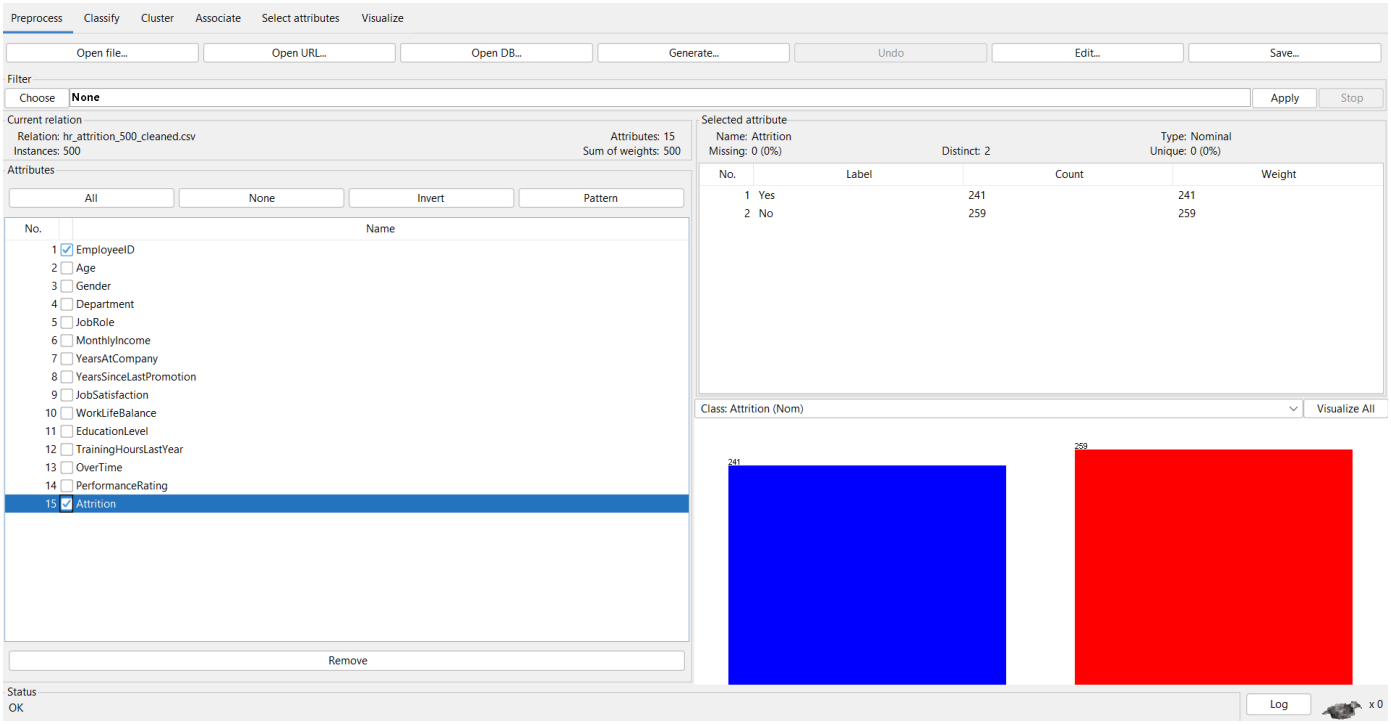
- Screenshot of ARFF Viewer with attributes visible



No.	1: EmployeeID Numeric	2: Age Numeric	3: Gender Nominal	4: Department Nominal	5: JobRole Nominal	6: MonthlyIncome Numeric	7: YearsAtCompany Numeric	8: YearsSinceLastPromotion Numeric	9: JobSatisfaction Numeric	10: WorkLifeBalance Numeric	11: EducationLevel Numeric	12: TrainingHoursLastYear Numeric	13: OverTime Nominal	14: PerformanceRating Numeric	15: **Attrition** Nominal
1	1001.0	50.0	Female	IT	Software ...	118694.0	10.0	7.0	1.0	1.0	2.0	74.0	Yes	3.0	Yes
2	1002.0	45.0	Male	IT	DevOps E...	141324.0	3.0	7.0	4.0	3.0	5.0	1.0	No	4.0	No
3	1003.0	51.0	Male	IT	Software ...	20311.0	20.0	0.0	4.0	2.0	5.0	48.0	Yes	3.0	Yes
4	1004.0	49.0	Male	R&D	Lab Tech...	17747.0	15.0	2.0	4.0	3.0	4.0	63.0	No	1.0	No
5	1005.0	24.0	Female	IT	Software ...	84092.0	18.0	6.0	1.0	1.0	2.0	3.0	No	1.0	No
6	1006.0	35.0	Female	Sales	Sales Exe...	83148.0	1.0	3.0	4.0	3.0	4.0	70.0	Yes	4.0	Yes
7	1007.0	29.0	Female	Finance	Accounta...	125510.0	2.0	0.0	4.0	2.0	4.0	1.0	No	2.0	No
8	1008.0	25.0	Female	IT	DevOps E...	24692.0	17.0	9.0	4.0	2.0	2.0	61.0	Yes	4.0	No
9	1009.0	35.0	Female	IT	DevOps E...	26534.0	7.0	6.0	4.0	1.0	5.0	79.0	No	5.0	No
10	1010.0	39.0	Male	IT	Software ...	97798.0	20.0	7.0	2.0	1.0	4.0	59.0	No	4.0	Yes
11	1011.0	50.0	Female	Sales	Sales Exe...	66214.0	12.0	0.0	1.0	3.0	1.0	0.0	No	4.0	No
12	1012.0	29.0	Female	HR	HR Execu...	139375.0	10.0	2.0	1.0	4.0	3.0	34.0	No	1.0	Yes
13	1013.0	26.0	Male	HR	HR Execu...	149633.0	6.0	9.0	1.0	4.0	1.0	71.0	Yes	4.0	Yes
14	1014.0	55.0	Male	Finance	Financial ...	67256.0	15.0	6.0	3.0	2.0	5.0	43.0	No	3.0	Yes
15	1015.0	56.0	Male	IT	Software ...	128252.0	14.0	2.0	1.0	1.0	2.0	13.0	No	3.0	No
16	1016.0	30.0	Male	HR	HR Execu...	146278.0	14.0	9.0	2.0	4.0	5.0	50.0	No	1.0	Yes
17	1017.0	53.0	Female	Finance	Financial ...	102939.0	4.0	6.0	3.0	1.0	3.0	28.0	Yes	4.0	No
18	1018.0	34.0	Female	Sales	Sales Exe...	77623.0	6.0	10.0	3.0	2.0	4.0	65.0	No	2.0	Yes
19	1019.0	27.0	Male	Finance	Financial ...	110259.0	11.0	3.0	2.0	2.0	3.0	61.0	Yes	1.0	Yes
20	1020.0	22.0	Male	IT	Software ...	143376.0	19.0	8.0	3.0	3.0	3.0	8.0	Yes	5.0	Yes
21	1021.0	54.0	Female	Finance	Accounta...	28545.0	11.0	7.0	3.0	3.0	1.0	51.0	No	1.0	No
22	1022.0	33.0	Male	Finance	Financial ...	67733.0	6.0	1.0	3.0	1.0	5.0	55.0	Yes	1.0	No
23	1023.0	23.0	Female	HR	HR Execu...	127859.0	4.0	0.0	1.0	3.0	2.0	52.0	No	4.0	Yes
24	1024.0	47.0	Female	Finance	Accounta...	85467.0	10.0	7.0	3.0	1.0	5.0	69.0	Yes	3.0	Yes
25	1025.0	37.0	Male	HR	HR Execu...	108384.0	5.0	2.0	4.0	4.0	3.0	35.0	No	3.0	No
26	1026.0	47.0	Female	R&D	Research ...	49754.0	18.0	3.0	4.0	3.0	4.0	32.0	No	2.0	No
27	1027.0	22.0	Male	HR	HR Mana...	147874.0	17.0	6.0	3.0	2.0	2.0	6.0	Yes	4.0	Yes
28	1028.0	37.0	Male	R&D	Research ...	111601.0	15.0	8.0	4.0	4.0	1.0	1.0	No	1.0	Yes
29	1029.0	33.0	Male	R&D	Research ...	35932.0	4.0	10.0	3.0	1.0	1.0	18.0	No	3.0	Yes
30	1030.0	41.0	Female	Sales	Sales Man...	21295.0	0.0	7.0	4.0	3.0	1.0	71.0	Yes	4.0	Yes
31	1031.0	41.0	Female	HR	HR Mana...	141063.0	2.0	8.0	1.0	2.0	2.0	65.0	No	3.0	No
32	1032.0	37.0	Male	IT	Software ...	65108.0	8.0	3.0	1.0	4.0	1.0	13.0	Yes	5.0	Yes
33	1033.0	41.0	Male	R&D	Research ...	73871.0	7.0	6.0	3.0	1.0	1.0	47.0	Yes	4.0	Yes
34	1034.0	43.0	Male	IT	Software ...	34830.0	13.0	5.0	2.0	1.0	3.0	53.0	Yes	3.0	No
35	1035.0	48.0	Male	IT	DevOps E...	64377.0	20.0	0.0	4.0	3.0	1.0	68.0	No	4.0	No

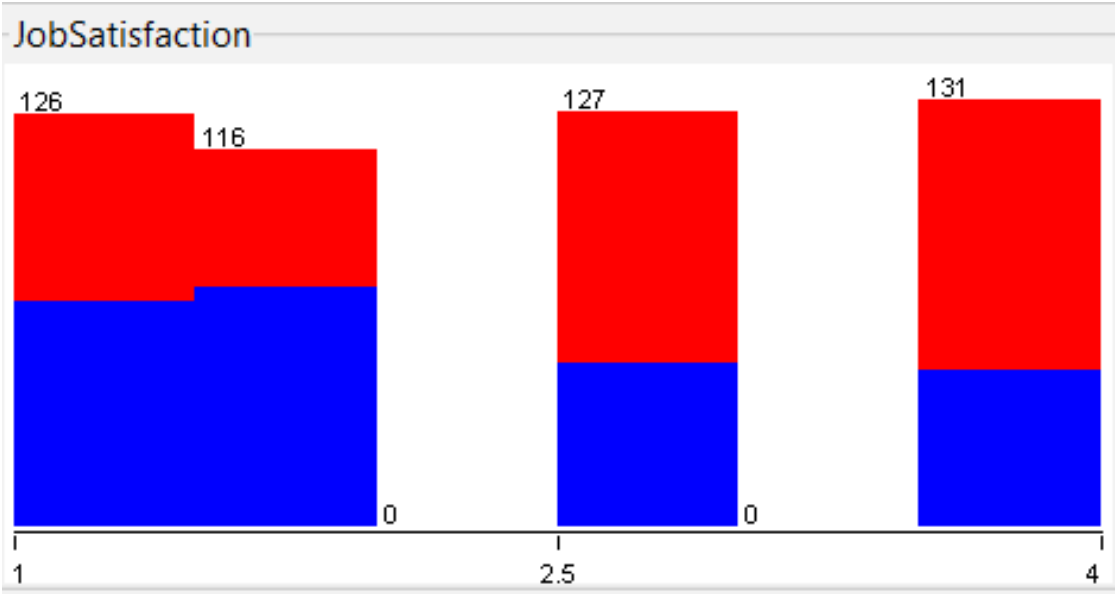
4. Load Dataset in Weka Explorer:

- Screenshot of attribute list & class variable

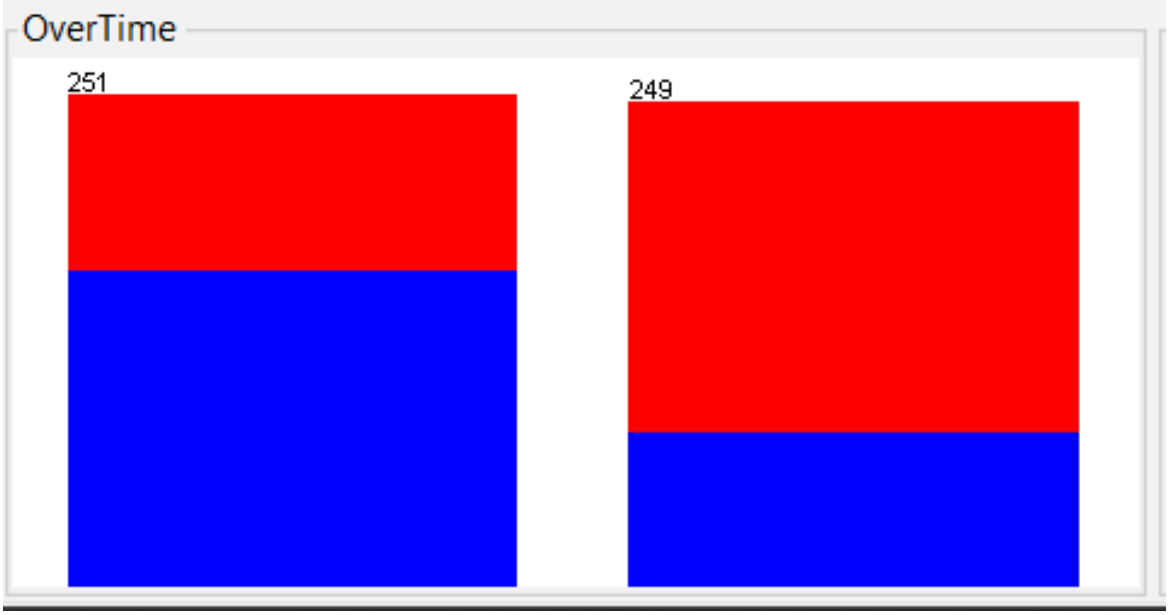


5. **<u>Explore Attributes & Visualize Patterns:</u>**

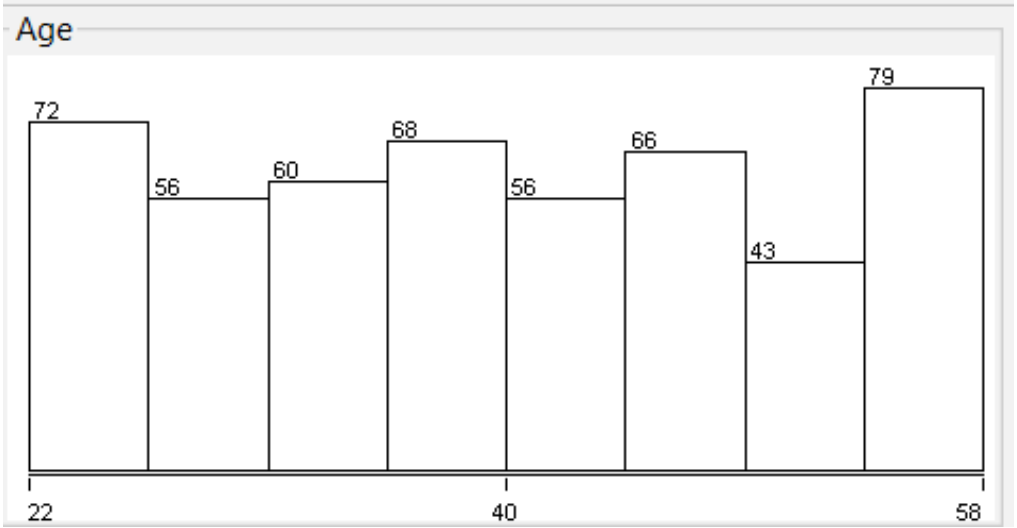
- Attrition vs Job Satisfaction

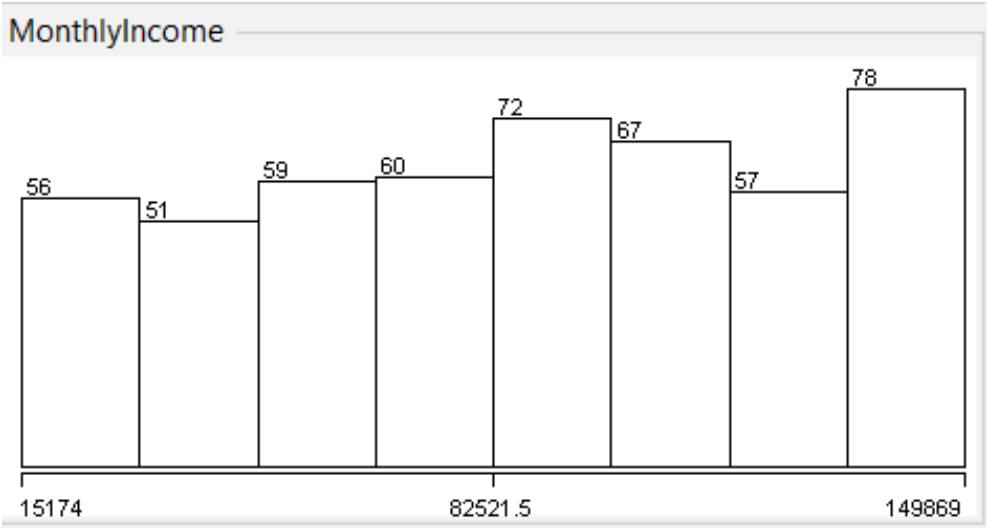


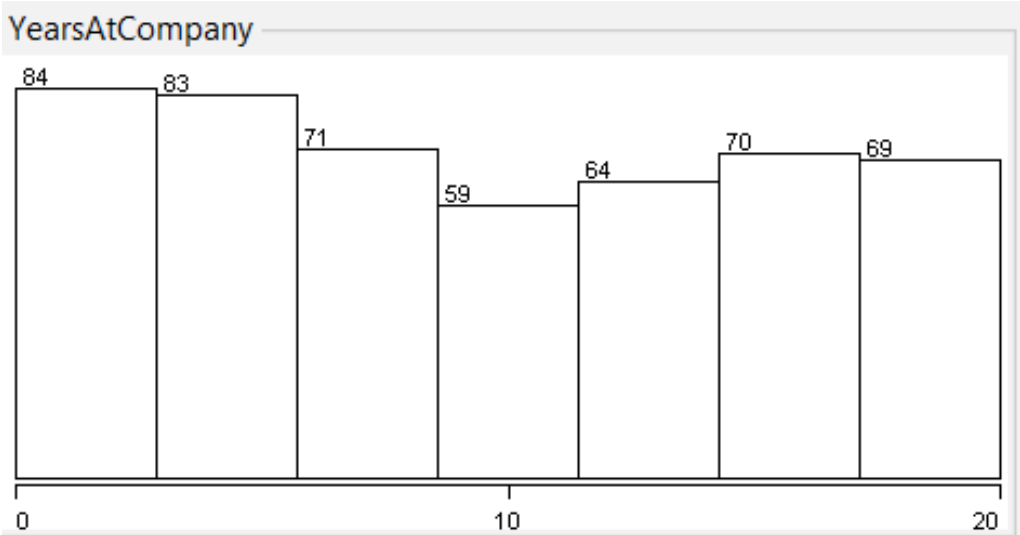
- Attrition vs OverTime



- Histogram:

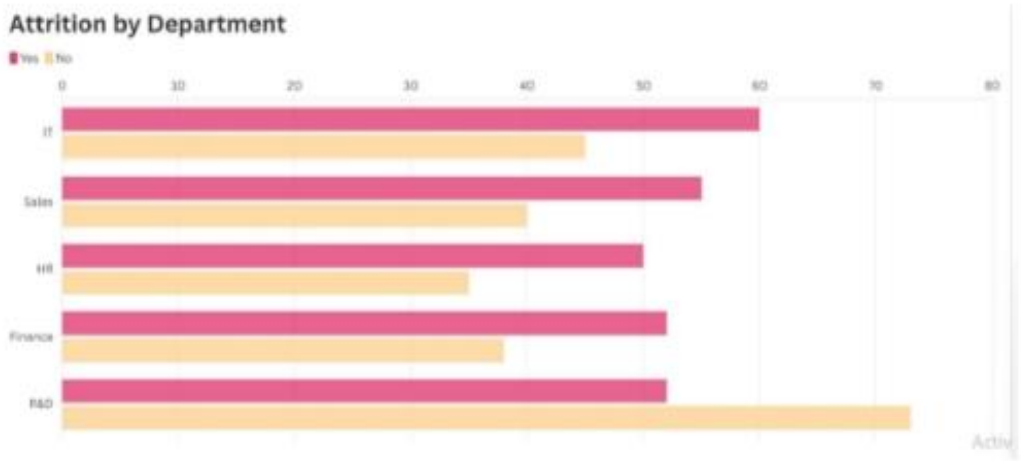

Age
72
56
60
68
56
66
43
79
22
40
58


MonthlyIncome
56
51
59
60
72
67
57
78
15174
82521.5
149869

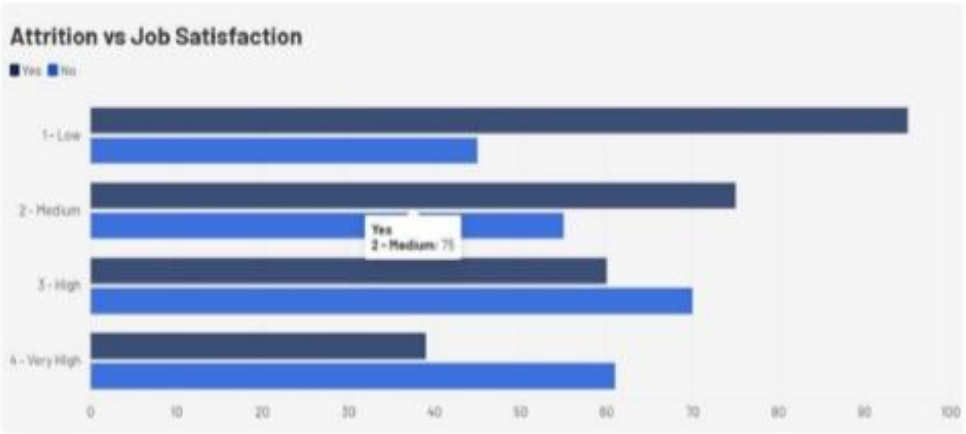

YearsAtCompany
84
83
71
59
64
70
69
0
10
20

6. Create Visual Narratives in Flourish:

- Attrition by Department (Bar Chart)



- Attrition vs Job Satisfaction (Scatter/Bar)



7. Train Baseline Model in Weka:

- Screenshot of classifier output
- Accuracy & confusion matrix

```
=== Summary ===

Correctly Classified Instances         433               86.6    %
Incorrectly Classified Instances        67               13.4    %
Kappa statistic                          0.7323
Mean absolute error                      0.2104
Root mean squared error                  0.3244
Relative absolute error                 42.1412 %
Root relative squared error             64.9164 %
Total Number of Instances              500

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.896    0.162    0.837      0.896    0.866      0.734    0.919     0.888     Yes
                 0.838    0.104    0.897      0.838    0.866      0.734    0.919     0.918     No
Weighted Avg.    0.866    0.132    0.868      0.866    0.866      0.734    0.919     0.904

=== Confusion Matrix ===

   a   b   <-- classified as
 216  25 |   a = Yes
  42 217 |   b = No
```

8. Overfitting Activity:

- Training set accuracy screenshot

```
=== Summary ===

Correctly Classified Instances         500               100      %
Incorrectly Classified Instances         0                 0      %
Kappa statistic                          1
Mean absolute error                      0
Root mean squared error                  0
Relative absolute error                  0      %
Root relative squared error              0      %
Total Number of Instances              500

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 1.000    0.000    1.000      1.000    1.000      1.000    1.000     1.000     Yes
                 1.000    0.000    1.000      1.000    1.000      1.000    1.000     1.000     No
Weighted Avg.    1.000    0.000    1.000      1.000    1.000      1.000    1.000     1.000

=== Confusion Matrix ===

   a   b   <-- classified as
 241   0 |   a = Yes
   0 259 |   b = No
```

- Cross-validation accuracy screenshot

```
=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances         310               62      %
Incorrectly Classified Instances       190               38      %
Kappa statistic                          0.2373
Mean absolute error                      0.3797
Root mean squared error                  0.616
Relative absolute error                 76.0359 %
Root relative squared error            123.2746 %
Total Number of Instances              500

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.573    0.336    0.613      0.573    0.592      0.238    0.620     0.558     Yes
                 0.664    0.427    0.625      0.664    0.644      0.238    0.620     0.591     No
Weighted Avg.    0.620    0.383    0.620      0.620    0.619      0.238    0.620     0.575

=== Confusion Matrix ===

   a   b   <-- classified as
 138 103 |   a = Yes
  87 172 |   b = No
```

9. Underfitting Activity:

- ZeroR model output
- Evaluation results (low accuracy)

```
=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances         259               51.8    %
Incorrectly Classified Instances       241               48.2    %
Kappa statistic                          0
Mean absolute error                      0.4994
Root mean squared error                  0.4997
Relative absolute error                100      %
Root relative squared error            100      %
Total Number of Instances              500

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.000    0.000    ?          0.000    ?          ?        0.496     0.480     Yes
                 1.000    1.000    0.518      1.000    0.682      ?        0.496     0.516     No
Weighted Avg.    0.518    0.518    ?          0.518    ?          ?        0.496     0.499

=== Confusion Matrix ===

   a   b   <-- classified as
   0 241 |   a = Yes
   0 259 |   b = No
```

10. **Summarize Findings:**

- Screenshot of Napkin AI summary

Model Evaluation and Insights

This document summarizes the performance of various classification models applied to a dataset of 500 instances. The models evaluated include J48 decision trees, an unpruned version of J48, stratified cross-validation, and a baseline ZeroR model. Each model's accuracy, error metrics, and confusion matrices are analyzed to derive insights and suggest potential improvements.

Model Performance Overview

J48 Decision Tree

- **Correctly Classified Instances:** 433 (86.6%)
- **Incorrectly Classified Instances:** 67 (13.4%)
- **Kappa Statistic:** 0.7323
- **Mean Absolute Error:** 0.2104
- **Root Mean Squared Error:** 0.3244
- **Total Instances:** 500

Detailed Accuracy by Class

Class	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area
Yes	0.896	0.162	0.837	0.896	0.866	0.734	0.919	0.888
No	0.838	0.104	0.897	0.838	0.866	0.734	0.919	0.918

Confusion Matrix

a	b	
216	25	a = Yes
42	217	b = No

Unpruned J48 Decision Tree

- **Correctly Classified Instances:** 500 (100%)
- **Incorrectly Classified Instances:** 0 (0%)
- **Kappa Statistic:** 1
- **Mean Absolute Error:** 0
- **Root Mean Squared Error:** 0

Detailed Accuracy by Class

Class	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area
Yes	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000
No	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000

Confusion Matrix

a	b	
241	0	a = Yes
0	259	b = No

Stratified Cross-Validation

- **Correctly Classified Instances:** 310 (62%)
- **Incorrectly Classified Instances:** 190 (38%)
- **Kappa Statistic:** 0.2373
- **Mean Absolute Error:** 0.3797
- **Root Mean Squared Error:** 0.616

Detailed Accuracy by Class

Class	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area
Yes	0.573	0.336	0.613	0.573	0.592	0.238	0.620	0.558
No	0.664	0.427	0.625	0.664	0.644	0.238	0.620	0.591

Confusion Matrix

a	b	
138	103	a = Yes
87	172	b = No

ZeroR Model

- **Correctly Classified Instances:** 259 (51.8%)
- **Incorrectly Classified Instances:** 241 (48.2%)
- **Kappa Statistic:** 0
- **Mean Absolute Error:** 0.4994
- **Root Mean Squared Error:** 0.4997

Detailed Accuracy by Class

Class	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area
Yes	0.000	0.000	?	0.000	?	?	0.500	0.482
No	1.000	1.000	0.518	1.000	0.682	?	0.500	0.518

Confusion Matrix

a	b	
0	241	a = Yes
0	259	b = No

Key Insights

1. **Model Performance:** The unpruned J48 model achieved perfect accuracy, but this may indicate overfitting, especially given the training set size. The standard J48 model performed well with an accuracy of 86.6%, while the stratified cross-validation showed a significant drop to 62%, indicating potential issues with generalization.
2. **Error Metrics:** The Kappa statistic for the J48 model indicates a strong agreement between predicted and actual classifications. However, the stratified cross-validation's Kappa statistic of 0.2373 suggests a weak agreement, highlighting the need for further model tuning.
3. **Class Imbalance:** The confusion matrices reveal a potential class imbalance issue, particularly in the stratified cross-validation results, where the model struggled to classify the "Yes" class accurately.
4. **Baseline Comparison:** The ZeroR model, which predicts the majority class, performed poorly overall but serves as a baseline for comparison. Its performance highlights the importance of model selection and tuning.

Suggested Improvements

1. **Hyperparameter Tuning:** Experiment with different parameters for the J48 model, such as pruning options and minimum object settings, to enhance generalization.
2. **Feature Engineering:** Investigate the features used in the models. Adding or transforming features may improve model performance.
3. **Class Balancing:** Implement techniques such as oversampling the minority class or undersampling the majority class to address class imbalance.
4. **Ensemble Methods:** Consider using ensemble methods like Random Forest or Gradient Boosting, which may provide better performance and robustness compared to single classifiers.
5. **Cross-Validation:** Increase the number of folds in cross-validation to better assess model performance and reduce variance in accuracy estimates.

By addressing these areas, the overall predictive performance of the models can be improved, leading to more reliable classifications in future applications.

- Findings From Napkin AI:

Model Evaluation and Insights

This document summarizes the performance of various classification models applied to a dataset of 500 instances. The models evaluated include J48 decision trees, an unpruned version of J48, stratified cross-validation, and a baseline ZeroR model. Each model's accuracy, error metrics, and confusion matrices are analyzed to derive insights and suggest potential improvements.

Model Performance Overview

J48 Decision Tree

- **Correctly Classified Instances**: 433 (86.6%)
- **Incorrectly Classified Instances**: 67 (13.4%)
- **Kappa Statistic**: 0.7323
- **Mean Absolute Error**: 0.2104
- **Root Mean Squared Error**: 0.3244
- **Total Instances**: 500

Detailed Accuracy by Class

| Class | TP Rate | FP Rate | Precision | Recall | F-Measure | MCC | ROC Area | PRC Area |

|-------|---------|---------|-----------|--------|-----------|-----|----------|----------|

| Yes | 0.896 | 0.162 | 0.837 | 0.896 | 0.866 | 0.734 | 0.919 | 0.888 |

| No | 0.838 | 0.104 | 0.897 | 0.838 | 0.866 | 0.734 | 0.919 | 0.918 |

Confusion Matrix

| a | b |

|-------|-------|

| 216 | 25 | a = Yes |

| 42 | 217 | b = No |

Unpruned J48 Decision Tree

- **Correctly Classified Instances**: 500 (100%)
- **Incorrectly Classified Instances**: 0 (0%)
- **Kappa Statistic**: 1
- **Mean Absolute Error**: 0
- **Root Mean Squared Error**: 0

Detailed Accuracy by Class

| Class | TP Rate | FP Rate | Precision | Recall | F-Measure | MCC | ROC Area | PRC Area |

|-------|---------|---------|-----------|--------|-----------|-----|----------|----------|

| Yes | 1.000 | 0.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |

| No | 1.000 | 0.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |

Confusion Matrix

| a | b |

|-------|-------|

| 241 | 0 | a = Yes |

| 0 | 259 | b = No |

Stratified Cross-Validation

- **Correctly Classified Instances**: 310 (62%)
- **Incorrectly Classified Instances**: 190 (38%)
- **Kappa Statistic**: 0.2373
- **Mean Absolute Error**: 0.3797
- **Root Mean Squared Error**: 0.616

Detailed Accuracy by Class

| Class | TP Rate | FP Rate | Precision | Recall | F-Measure | MCC | ROC Area | PRC Area |

|-------|---------|---------|-----------|--------|-----------|-----|----------|----------|

| Yes | 0.573 | 0.336 | 0.613 | 0.573 | 0.592 | 0.238 | 0.620 | 0.558 |

| No | 0.664 | 0.427 | 0.625 | 0.664 | 0.644 | 0.238 | 0.620 | 0.591 |

Confusion Matrix

| a | b |

|-------|-------|

| 138 | 103 | a = Yes |

| 87 | 172 | b = No |

ZeroR Model

- **Correctly Classified Instances**: 259 (51.8%)
- **Incorrectly Classified Instances**: 241 (48.2%)
- **Kappa Statistic**: 0
- **Mean Absolute Error**: 0.4994
- **Root Mean Squared Error**: 0.4997

Detailed Accuracy by Class

| Class | TP Rate | FP Rate | Precision | Recall | F-Measure | MCC | ROC Area | PRC Area |

|-------|---------|---------|-----------|--------|-----------|-----|----------|----------|

| Yes | 0.000 | 0.000 | ? | 0.000 | ? | ? | 0.500 | 0.482 |

| No | 1.000 | 1.000 | 0.518 | 1.000 | 0.682 | ? | 0.500 | 0.518 |

Confusion Matrix

| a | b |

|-------|-------|

| 0 | 241 | a = Yes |

| 0 | 259 | b = No |

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<u>Reflection Questions:</u>

1. Which attributes were most important for attrition?

Answer: The most important factors were job satisfaction and overtime. Employees with low job satisfaction and those who worked a lot of overtime were more likely to leave. Work-life balance also played a role.

2. Was your model overfitting or underfitting? Why?

Answer: The model was overfitting. This is because the unpruned J48 model gave 100% accuracy on the training data, but the accuracy dropped significantly to around 62% during cross-validation. This shows that the model memorized the training data instead of learning general patterns, so it does not perform well on new, unseen data.

3. What data improvements could reduce errors?
Answer: Errors could be reduced by adding more data, fixing class imbalance, and improving data quality. Including more useful features like salary, promotions, or workload could also help.

4. How should HR teams use this model in decision-making?
Answer: HR teams can use this model to identify employees who are likely to leave. Based on this, they can take early actions like improving work conditions, offering incentives, or reducing workload to retain employees.

End Report